Segmenting Objects in Day and Night: Edge-Conditioned CNN for Thermal Image Semantic Segmentation

Chenglong Li, Wei Xia, Yan Yan, *Member, IEEE*, Bin Luo, *Senior Member, IEEE*, and Jin Tang

***Abstract*— Despite much research progress in image semantic segmentation, it remains challenging under adverse environmental conditions caused by imaging limitations of the visible spectrum, while thermal infrared cameras have several advantages over cameras for the visible spectrum, such as operating in total darkness, insensitive to illumination variations, robust to shadow effects, and strong ability to penetrate haze and smog. These advantages of thermal infrared cameras make the segmentation of semantic objects in day and night. In this article, we propose a novel network architecture, called edge-conditioned convolutional neural network (EC-CNN), for thermal image semantic segmentation. Particularly, we elaborately design a gated featurewise transform layer in EC-CNN to adaptively incorporate edge prior knowledge. The whole EC-CNN is end-to-end trained and can generate high-quality segmentation results with edge guidance. Meanwhile, we also introduce a new benchmark data set named "Segmenting Objects in Day And night" (SODA) for comprehensive evaluations in thermal image semantic segmentation. SODA contains over 7168 manually annotated and synthetically generated thermal images with 20 semantic region labels and from a broad range of viewpoints and scene complexities. Extensive experiments on SODA demonstrate the effectiveness of the proposed EC-CNN against state-of-the-art methods.**

***Index Terms*— Benchmark data set, edge prior knowledge, featurewise affine transform, semantic segmentation, thermal infrared image.**

I. Introduction

THE task of image semantic segmentation is to assign a semantic label to every pixel in an image. It is a high-level task that paves the way toward scene understanding, whose importance is highlighted by the fact that an increasing number of applications nourish from inferring knowledge from imagery, including autonomous driving, computational photography, and augmented reality, to name a few. Although remarkable progress has been achieved in the past decades, image semantic segmentation is still considered to be challenging due to the imaging limitations of visible cameras. For example, visible cameras are very sensitive to lighting conditions and become invalid in total darkness. Moreover, their imaging quality decreases significantly in adverse environmental conditions, such as rain, haze, and smog.

Thermal infrared cameras can capture infrared radiation (0.75–13 μm) emitted by subjects with a temperature above absolute zero. Therefore, the thermal imaging reflects the temperature distribution of a scene and has many advantages over visible ones, such as insensitivity to lighting conditions and strong penetration to smog and fog. There are many applications of thermal cameras used for computer vision tasks, including the RGBT/thermal object tracking [2]–[5], multispectral pedestrian detection [6], [7], and multispectral vehicle Re-ID [8]–[10].

Given the potentials of thermal data, it is interesting to investigate the problem of semantic segmentation on thermal images, called thermal image semantic segmentation in this article. The thermal image semantic segmentation is similar to the general task of semantic image segmentation as their goal is the same as to classify each pixel, while the task of thermal image semantic segmentation has some different challenges and properties. For example, two subjects with similar temperatures are difficult to distinguish since the thermal image reflects the temperature distribution of a scene. Benefiting from the aforementioned advantages of thermal cameras, thermal image semantic segmentation has wider applications, such as electrical equipment diagnosis, forest fire warning, and UAV surveillance.

However, what is a reasonable baseline algorithm for thermal image semantic segmentation? How to construct a reasonable size and annotated thermal image benchmark for persuasive comparisons of different algorithms? This article seeks to answer these questions by proposing a powerful baseline algorithm for thermal semantic segmentation, creating a comprehensive and elaborately annotated thermal image data set, and conducting a quantitative benchmark evaluation for different baseline algorithms. First, we propose a novel network architecture, called edge-conditioned convolutional

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TABLE I
COMPARE WITH POPULAR LARGE-SCALE SEGMENTATION DATA SETS

Name and Reference	Classes	Visible/Depth/Thermal	Resolution	Samples(training)	Samples(validation)	Samples(test)	Samples(total)
PASCAL VOC 2012 Segmentation [40]	20	V	Variable	1,464	1,449	Private	2,913+
PASCAL-Context [41]	540	V	Variable	10,103	N/A	9,637	19,740
PASCAL-Part [42]	20	V	Variable	10,103	N/A	9,637	19,740
Microsoft COCO [43]	80	V	Variable	82,783	40,504	81,434	204,721
Cityscapes(fine) [44]	30(8)	V	2048x1024	2975	500	1525	5,000
CamVid [45]	32	V	960x720	701	N/A	N/A	701
KITTI-Layout [46, 47]	3	V	Variable	323	N/A	N/A	323
SiftFLow [48]	33	V	256x256	2,688	N/A	N/A	2,688
NYUDv2 [49]	40	V+D	480x640	795	654	N/A	1449
SUNRGBD [50]	37	V+D	Variable	2,666	2,619	5,050	10,335
RGB-D Object Dataset [51]	51	V+D	640x480	207,920	N/A	N/A	207,903
SODA(+Synthetic Cityscapes fine)	20	T	640x480	1,168+5,000	N/A	1,000	7,168

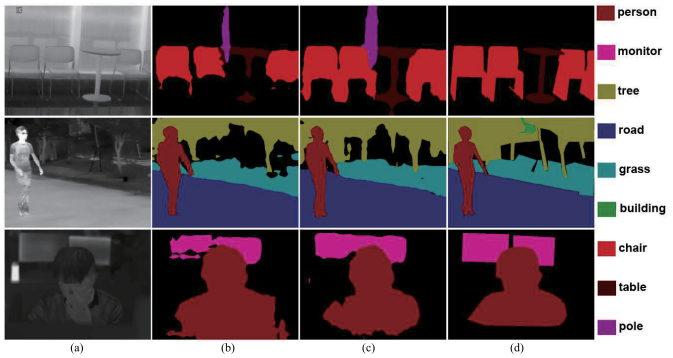


Fig. 1. Three examples for demonstrating the effectiveness of the proposed EC-CNN. (a) Input thermal images. (b) Results generated by the state-of-the-art network DeepLabv3 [1]. (c) Results generated by our EC-CNN. (d) Ground-truth annotations.

neural network (EC-CNN), for thermal image semantic segmentation. Despite many advantages on visible spectrum images [1], [11], thermal images are usually with low resolution and contrast, ambiguous object boundaries caused by thermal crossover, and imaging noised introduced by thermal sensors. Therefore, directly adapting existing techniques for thermal image semantic segmentation is difficult to distinguish between different semantic objects accurately, as shown in Fig. 1(a) and (b). To handle these problems, we employ the edge prior knowledge as weakly supervised information to guide thermal image semantic segmentation, and the results are demonstrated in Fig. 1(c). In particular, based on the network DeepLabv3 [1], we elaborately design the gated featurewise transform (GFT) layer in EC-CNN to adaptively embed edge information as the guidance of semantic segmentation. The major advantages of GFT are as follows: 1) it is parameter-efficient as just a single forward pass through a single network conditioned on the edge prior is needed; 2) it is able to alleviate the effects of noisy edge information via the designed gate scheme; 3) it is able to generate high-quality segmentation results through the adaptive guidance of the edge probability maps; and 4) it is generic as it could be applied to other network structures of semantic segmentation. Note that we train our EC-CNN in an end-to-end manner.

Second, we establish a standard thermal image benchmark for semantic segmentation. There is no public data set for thermal image semantic segmentation (see Table I for details), and we contribute such a benchmark data set to facilitate the research. Taking advantage of thermal imaging, we are able to segment any object in day and night, and thus, we call our benchmark data set Segmenting Objects in Day And night (SODA) for convenience. SODA contains more than 2000 manually annotated and 5000 synthetically generated thermal images with 20 semantic region labels and covers a broad range of viewpoints and scene complexities. Particularly, to save the tedious pixelwise annotation cost, we annotate a small set of thermal images and generate a large set of thermal images using the image-to-image technique [3], [12].

Finally, we conduct extensive experiments on the SODA data set. To show the advance of our EC-CNN, we include 11 state-of-the-art methods of image semantic segmentation and three evaluation metrics (i.e., mean IoU, pixel accuracy (pixAcc), and runtime) in our SODA benchmark for persuasive comparisons of different algorithms. With the deep analysis of these comparison results, we provide some insights and potential future research directions for thermal image segmentation.

To the best of our knowledge, it is the first work to investigate the problem of thermal image semantic segmentation. Our contributions are summed up as follows.

1) We propose a novel neural network EC-CNN for thermal image semantic segmentation. The EC-CNN is able to adaptively embed the edge priors and is, thus, able to boost the quality of segmentation results.
2) To embed edge prior knowledge effectively, we design a GFT layer to incorporate hierarchical edge maps in an adaptive manner. The designed layer can perform effective edge embedding while alleviating effects of noises introduced by the edge extraction network.
3) We create a new image benchmark data set SODA for thermal image semantic segmentation. It provides a first evaluation platform for thermal semantic segmentation and would be available online for free academic usage.[1]
4) Extensive experiments on the SODA data set are carried out to demonstrate the effectiveness of the proposed EC-CNN. Through analysis of experimental results, we provide some insights and potential research directions to the field of thermal image semantic segmentation.

[1]Thermal image semantic segmentation data set: http://chenglongli.cn/people/lcl/dataset-code.html

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II. Related Work

This section reviews the most relevant works to ours, including semantic segmentation networks, conditioned networks, and adversarial image-to-image translation.

A. Image Semantic Segmentation

The most successful state-of-the-art deep learning techniques for semantic segmentation stem from a common forerunner, the fully convolutional network (FCN) [13]. FCN replaces the later fully connected layers with convolutional ones to output fully resolution maps instead of classification scores and is the first work to show how CNNs can be trained end-to-end for this problem. Based on FCN, several types of improvements have been made recently.

The encoder–decoder architectures for semantic segmentation [14], [15] first encode longer range information with the spatial dimension gradually reduced and then decode to recover object details and spatial resolution. For example, SegNet [14] learns extra convolutional layers to densify the feature response. To encode long-range context while preserving object details, four typical methods, including image pyramid [11], [16], CRF [11], spatial pyramid pooling [11], and atrous convolution [1], [17], are developed. Chen *et al.* [11], [16] directly resize the input for several scales and fuse the features from all scales. To overcome the poor localization property of CNNs, the DenseCRF [18] is combined with the responses at the final CNN layer [11]. The atrous spatial pyramid pooling is used in DeepLabv2 [11] to capture multiscale information via equipping parallel atrous convolutional layers. Chen *et al.* [1] design modules that employ atrous convolution in cascade or in parallel to capture multiscale context by adopting multiple atrous rates, while Wang *et al.* [17] adopt the atrous convolution with hybrid atrous rates within the last two blocks of ResNet for capturing long-range information. Yu *et al.* [19] adopt encoder–decoder structures that fuse the information in low- and high-level layers to predict segmentation mask. Li *et al.* [20] use a lightweight backbone and equips with sub-stage aggregation for real-time semantic segmentation.

Besides, there are some works [21]–[23] to use edge cues for image semantic segmentation [21]–[23]. Kokkinos [21] designs the CRF-based method, which integrates CNN score-maps with pairwise features derived from advanced boundary detection, to more precisely localize semantic segmentation boundaries. The outputs of the edge network are employed to refine the semantic segmentation network by adding an edge-aware regularization so as to yield spatially consistent and well boundary located results [23]. Chen *et al.* [22] uses the DeepLap network to simultaneously predict edge map and segmentation scores and then utilizes the domain transform to align segmentation scores with edge prediction. Although we adopt the DeepLab variant as the baseline, our method differs it from the following aspects. First, we predict edge results using an additional network, which could extract object boundaries more accurately. Second, we integrate edge features instead of final edge prediction into the segmentation network, and more, contextual edge information could, thus, be incorporated. Third, we use a GFT module to embed edge priors more robust and accurate.

B. Conditioned Networks

Conditional feature normalization uses a learned function of some conditions to achieve featurewise affine transformation in batch normalization (BN) [24], and its variants are proven to be effective in several computer vision tasks, such as style transfer [25], [26], visual question answering [27], visual reasoning [28], and image super-resolution [29]. Perez *et al.* [28] propose a featurewise linear modulation layer (FiLM) to exploit linguistic information for visual reasoning. For the low-level tasks, such as image super-resolution and semantic segmentation, the spatial information is crucial and, thus, should be considered in feature modulation. Wang *et al.* [29] propose spatial feature transform (SFT) to incorporate semantic prior for image super-resolution. However, noisy information introduced by the priors might affect the performance. Therefore, we design a gate scheme in the SFT to adaptively incorporate edge priors in our framework.

C. Adversarial Image-to-Image Translation

Generative adversarial networks (GANs) [30] have achieved promising results in several task, such as image generation [31], image inpainting [32], and representation learning [33]. The conditional variants of GANs [34] enable to condition of the image generation on a selected input variable, for example, an input image. In this case, the task becomes image-to-image translation, and this is the variant we use here. The general method of Isola *et al.* [35], pix2pix, was the first GAN-based image-to-image translation work that was not designed for a specific task. The architecture is based on an encoder–decoder with skip connections, and it is trained using a combination of losses: a conditional adversarial loss [30] and a loss that maps the generated image close to the corresponding target image. This method achieves excellent results but loss the details of object delineation, in other words, failure for high-resolution image generation tasks. In order to overcome this limitation, [12] extended this model to provide a new adversarial learning objective together with new multiscale generator and discriminator architectures, which are more visually appealing than those computed by previous methods. We use it to create synthetic training data for thermal image semantic segmentation; it is much easier to create semantic segmentation data sets for desired thermal image scenarios than to generate training images.

III. Edge-Conditioned CNN

To utilize the edge prior sufficiently from thermal images for semantic segmentation, we design an EC-CNN, and this section will present the details of our EC-CNN.

A. EC-CNN Architecture

As shown in Fig. 2, our EC-CNN consists of two streams, including an edge detection network and a novel semantic segmentation network.

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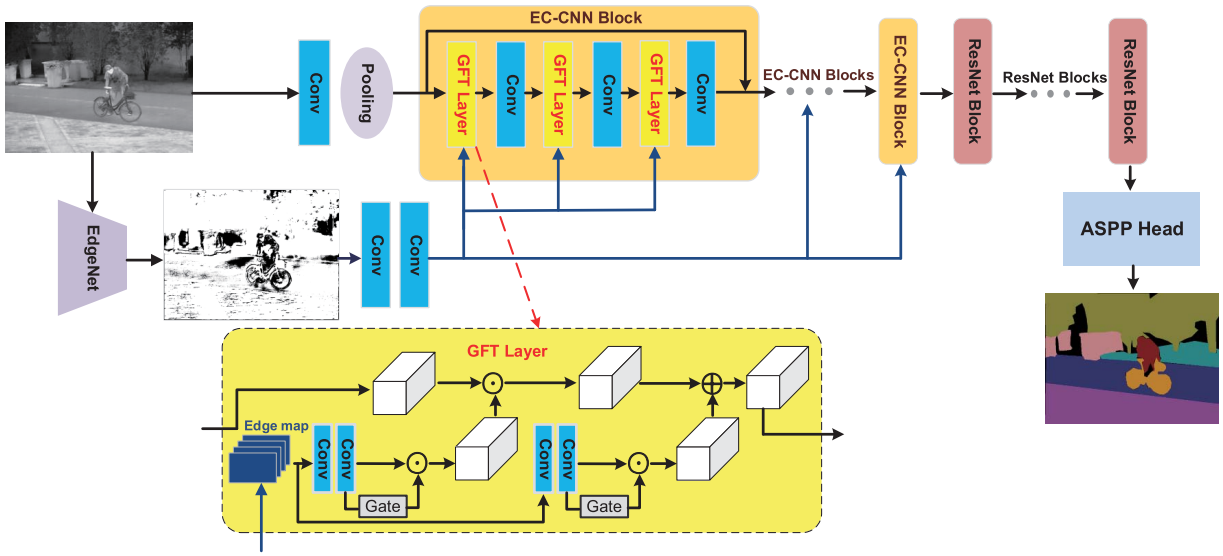


Fig. 2. Diagram of our EC-CNN architecture. The input image is first processed via an HED to generate hierarchical edge maps, which are embed into the EC-CNN block by inserting the GFT layers to guide the segmentation of the input image. Taking the DeepLabv3 as a baseline, we replace some ResNet blocks with EC-CNN blocks for edge prior embeddings.

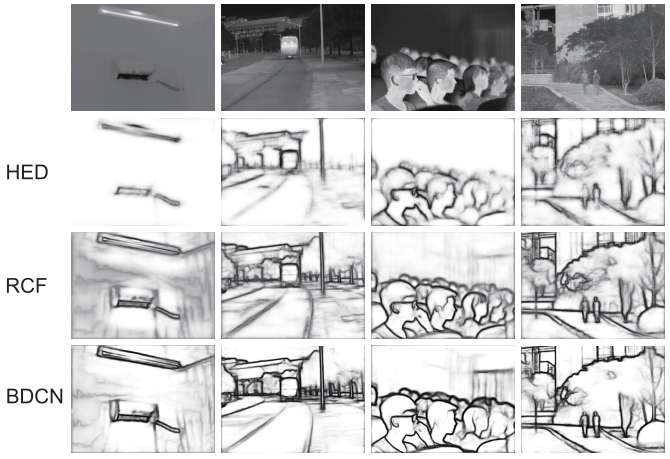


Fig. 3. Different edge results trained on RGB-based data set.

Despite many advantages on visible spectrum images [1], [11], thermal images are usually with low resolution and contrast, ambiguous object boundaries caused by thermal crossover, and imaging noised introduced by thermal sensors. To handle this problem, we aim to utilize hierarchical edge features to guide thermal image segmentation and, thus, equip the edge detection network in our network. These edge priors are beneficial for guiding the results of thermal semantic segmentation to possess good object boundaries. Especially, we employ the HED [36] as our edge detection network to generate high-quality edge features. Note that there is no thermal image data set with ground-truth edge annotations. Therefore, we only train HED using an RGB image data set, BSD500 [37]. In spite of using RGB images as training samples, the high-quality edge results of thermal images are also obtained mainly as RGB, and thermal images are able to share low-level feature representations, as shown in Fig. 3.

The proposed semantic segmentation network bases on a state-of-the-art semantic segmentation network, i.e., DeepLabv3 [1]. We briefly review the DeepLabv3 here for clarity. DeepLabv3 takes the structure of the ResNet [38] as its feature extractor. To enable different-level context aggregation and retain more spatial information, it integrates the dilated/atrous convolutions with different rates in the network. To handle the problem of segmenting objects at multiple scales, it employs atrous convolution in cascade or in parallel to capture multiscale context by adopting multiple atrous rates. An atrous spatial pyramid pooling module is incorporated to probe convolutional features at multiple scales, which encodes global context and further boosts performance.

To deal with the challenges of low resolution and contrast, and ambiguous object boundaries caused by thermal crossover, we embed edge feature maps generated by HED into DeepLabv3. To this end, we design an EC-CNN block by inserting some GFT layers, as shown in Fig. 2, and integrate it into DeepLabv3 so that the edge priors could be deployed to guide thermal image semantic segmentation. The GFT layer consists of three components, i.e., **feature modulation**, **information control**, and **featurewise transform**.

Since the edge maps generated by HED have different resolutions and channel numbers from the feature maps of DeepLabv3, we cannot utilize the edge priors directly. To handle this problem, we design a module of feature modulation to transform the resolution and channel number of edge maps into the same with the feature maps. The feature modulation includes two convolution layers and the hyperparameters of them are adaptively set according to the depth of feature maps. The edge prior information generated from HED often includes many noises, and direct use of them might degrade the performance. Therefore, we design an information control module using a gate scheme to adaptively incorporate edge prior information. As for the module of featurewise transform,

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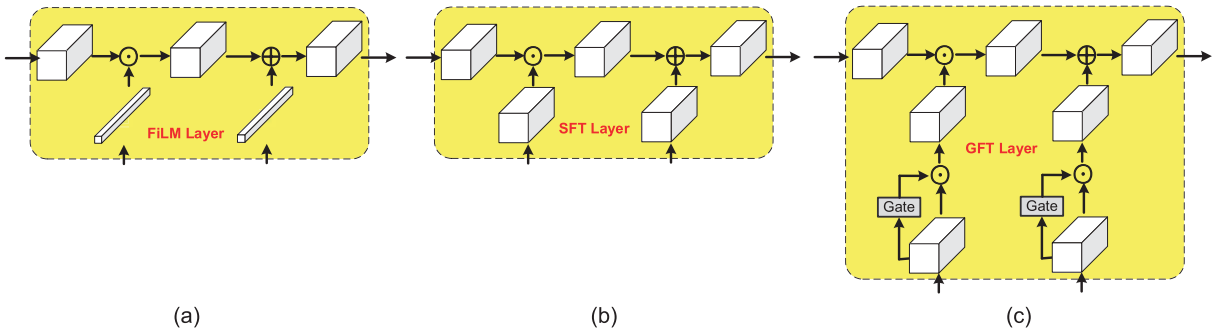


Fig. 4. Structures of (a) FiLM [28], (b) SFT [29], and (c) our GFT, respectively.

we use it to embed edge maps into the learning of deep features by applying a spatially affine transformation to each intermediate feature map in DeepLabv3. Next, we discuss the GFT layer in detail.

B. GFT Layer

Our EC-CNN is inspired by previous studies on feature normalization [24], [25]. BN is a widely used technique to ease network training by normalizing features statistics [24]. Conditional normalization (CN) applies a learned function of some conditions to replace parameters for featurewise affine transformation in BN. Some variants of CN have been proven to be highly effective in image style transfer [25], [26], visual question answering [27], and visual reasoning [39]. Perez *et al.* [28] develop an FiLM to exploit linguistic information for visual reasoning. FiLM is a relevant technique to us, and thus, we briefly introduce it first.

An FiLM layer applies a featurewise affine transformation (i.e., the shifting and scaling coefficients) to condition prior knowledge on feature maps for more robust feature learning. The structure of FiLM is shown in Fig. 4(a). We denote that $\mathbf{x}$ is an input of FiLM layer, $\mathbf{z}$ is a conditioning input, and γ and β are $\mathbf{z}$-dependent functions that represent the scaling and shifting vectors, respectively. The formulation of FiLM can be written as follows:

$$\mathrm{FiLM}(\mathbf{x}) = \gamma(\mathbf{z}) \times \mathbf{x} + \beta(\mathbf{z}) \tag{1}$$

where the dimensions of $\gamma(\mathbf{z})$ and $\beta(\mathbf{z})$ are equal to the channel number of $\mathbf{x}$, i.e., each channel of $\mathbf{z}$ is mapped into a scaling parameter and a shifting parameter.

Nonetheless, FiLM cannot handle the conditions with spatial information since it maps each channel of input maps into a single scaling/shifting parameter. The spatial information is crucial for low-level tasks, such as image super-resolution and semantic segmentation, since these tasks usually require adaptive processing at different spatial locations of an image. Therefore, Wang *et al.* [29] propose an SFT to incorporate spatial priors for image super-resolution. SFT is formulated as follows:

$$\mathrm{SFT}(\mathbf{x}) = \gamma(\mathbf{z}) \odot \mathbf{x} \oplus \beta(\mathbf{z}) \tag{2}$$

where $\odot$ and $\oplus$ denote the elementwise product and sum, respectively. Different from FiLM, the dimensions of $\gamma(\mathbf{z})$ and $\beta(\mathbf{z})$ are the same with $\mathbf{z}$, i.e., the input $\mathbf{z}$ is mapped into $\mathbf{z}$-dependent scaling and shifting tensors. The structure of SFT is shown in Fig. 4(b).

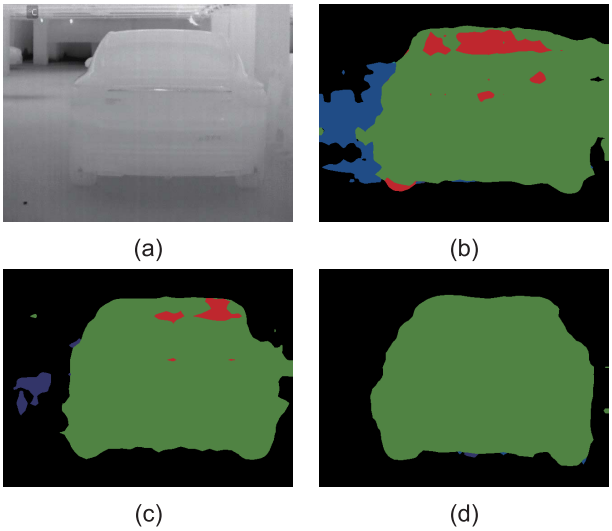


Fig. 5. Illustration of the proposed components. (a) Input image. (b) Result of DeepLabv3 [1]. (c) Result of our method without the gate scheme. (d) Result of our method. The results show the effectiveness of the components in our method.

We aim to employ SFT to embed the edge priors to guide thermal image semantic segmentation. However, the HED often introduces some noisy edges, and directly using these edges as guidance would affect the performance of semantic segmentation. To handle this problem, we propose a new layer called GFT to achieve the adaptive incorporation of edge priors. The effectiveness of the proposed GFT is demonstrated in Fig. 5. As discussed earlier, GFT includes three modules of feature modulation, information control, and featurewise transform. The information control module is used to adaptively incorporate edge prior information, and we propose to use a gate scheme to realize it. The structure of our GFT is shown in Fig. 4(c). Especially, we formulate the information control module as follows:

$$\begin{aligned}(\gamma^*(\hat{\mathbf{z}}), \beta^*(\hat{\mathbf{z}})) &= \sigma(\mathrm{Conv}(\gamma(\hat{\mathbf{z}})), \quad \mathrm{Conv}(\beta(\hat{\mathbf{z}}))) \\ \hat{\gamma}(\hat{\mathbf{z}}) &= \gamma(\hat{\mathbf{z}}) \odot \gamma^*(\hat{\mathbf{z}}) \\ \hat{\beta}(\hat{\mathbf{z}}) &= \beta(\hat{\mathbf{z}}) \odot \beta^*(\hat{\mathbf{z}})\end{aligned} \tag{3}$$

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where σ is the sigmoid function, and $\hat{\mathbf{z}}$ is the output of the feature modulation module with the input of $\mathbf{z}$. $Conv$ denotes the convolutional operation, and here, the kernel size is set to 1*1. Therefore, the GFT can be formulated as follows:

$$\mathrm{GFT}(\mathbf{x}) = \hat{\gamma}(\hat{\mathbf{z}}) \odot \mathbf{x} \oplus \hat{\beta}(\hat{\mathbf{z}}). \tag{4}$$

Fig. 2 shows the detailed structure of the GFT layer.

C. Loss Function

We apply the multiclass pixelwise cross entropy loss for semantic segmentation as follows:

$$\mathcal{L} = -\sum_{j}^{C}\sum_{i}^{N} \mathbf{y}_i^j \log G_\theta\left(\mathbf{x}_i^j\right) \tag{5}$$

where $G_\theta(\mathbf{x}_i^j)$ is the label assignment probability at the jth pixel in the ith class with the parameter θ in our EC-CNN, and $\mathbf{y}_i^j$ is the ground-truth label vector with the dimension of C. N is the number of pixels.

D. Network Training

As for HED, we first initialize its parameters using the pretrained VGG-16-Net model on the ImageNet data set and then fine-tuned it using the BSD500 data set [37] to generate hierarchical edge maps and keep its parameters unchanged in the training procedure of EC-CNN. Although the model is trained on RGB data, the high-quality edge results of thermal images are also obtained mainly as RGB and thermal images are able to share low-level feature representations as discussed earlier. To effectively learn the parameters of EC-CNN using our SODA data set, we adopt a progressive learning algorithm. First, we load the network parameters pretrained on the ImageNet data set as the initial parameters of EC-CNN. Second, we use the synthetic thermal subset to further train EC-CNN. Finally, we fine-tune our network by the real thermal subset. Note that the labels of the Cityscapes data set are different from our real subset; thus, we discard the parameters of the last layer after the second stage.

IV. SODA Data Set

This section will introduce the benchmark data set, namely, SODA. Particularly, our SODA includes 2168 annotated images and 5000 synthetically generated thermal images. The former is captured in real scenes and environments using a thermal camera, but the manual annotations are time-consuming and also easily include noises due to low contrast and resolution of thermal images. To handle these problems, we employ some well-annotated RGB images to generate synthetic thermal images. We first present the details of the real part of SODA and then the synthetic part.

A. Real Subset

Our SODA data set is captured under an FLIR thermal camera (SC620) from a broad range of viewpoints and scene complexities and contains 2168 elaborately annotated thermal images with 20 semantic region labels, including person, building, tree, road, pole, grass, door, table, chair, car, bicycle, lamp, monitor, trafficCone, trash can, animal, fence, sky, river, sidewalk, and a background label. The images collected from the real-world scenarios contain variety of appearances with different viewpoints, heavy occlusions, imaging blur, and in a wide range of resolutions.

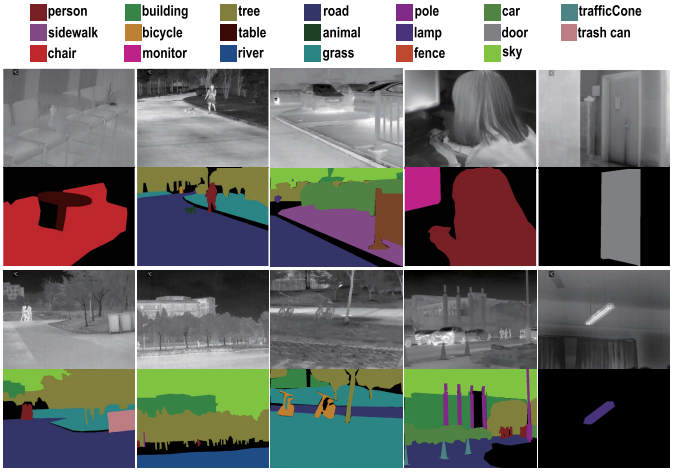


Fig. 6. Examples of thermal images and their corresponding ground-truth annotations from the SODA data set.

We use *labelme*, an open-source annotation tool, to annotate thermal images. First, we input an image, and click a number of pixels along the boundary of an object to form a closed polygon. Then, the polygon region is cropped and assigned with a predefined label. Finally, the annotation information (region boundary and class label) is added to the database.

We invite 12 annotation experts to elaborately annotate each image and check the annotated images periodically to keep the annotation quality. In particular, since thermal images are usually with low resolution and low contrast, there are many ambiguous regions that are hard to distinguish. If the annotators totally do not distinguish these regions, we treat them as background classes. If the annotators are able but difficult to distinguish their regions, we empirically annotate them. After annotated all images, we conduct a second-round check to remove nonstandard ones and select 2168 usable and well-annotated images from over 4000 submitted ones. Fig. 6 shows some examples of SODA.

For the better understanding of our data set, we analyze the statistics of the real part in the SODA data set. There are 20 categories divided into two types: 1) indoor items and 2) outdoor items. Indoor items include common categories such as chair, lamp, monitor, and table. Outdoor items include the tree, road, building, sky, car, grass, and so on. The characteristics of thermal images are often ambiguity and fuzzy compared with RGB images. In total, there are 2168 images, including 166 animal images, 384 bicycle images, 1013 building images, 769 car images, 148 chair images, 87 door images, 172 fence images, 752 grass images, 40 lamp images, 66 monitor images, 1197 person images, 537 pole images, 122 river images, 1248 road images, 818 sidewalk images, 835 sky images, 77 table images, 203 trafficCone images, 162 trash can images, and 1524 tree images.

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Fig. 7. Translate TIR of Cityscapes (top two) and NYUv2 (bottom two).

B. Synthetic Subset

As discussed earlier, we also construct a set of synthetic thermal images to save manual costs and improve the quality of annotations.

To this end, we use the pix2pixHD [12], an image-to-image translation method, to transform labeled RGB images into thermal images, and the labels of thermal images could directly be transferred from RGB ones. Among the paired data sets [2], [4], [54], the biggest and most relevant is KAIST Multispectral Pedestrian Data set [54], which contains a significant amount of aligned images in the RGB and thermal modalities, captured from a moving vehicle in different urban environments and under different lighting conditions. We follow the article [54] and use all the frames of training videos to train the pix2pixHD.

According to the relevance to the thermal images in our real subset, we select the Cityscapes [44], an RGB-based cityscapes data set for semantic urban scene understanding, to generate our synthetic subset. After acquiring a trained model, we translate Cityscapes [44] to thermal cityscapes, as shown in Fig. 7. The newly generated subset is made up of synthetic thermal images and the original semantic label with fine annotations.

C. Training and Testing Sets

Considering the scale of our SODA data set, we divide it into training and testing sets. Usually, the proportion of training and testing test sets are about 1:1 in semantic segmentation data set [41]–[43]. Therefore, we divide the manual-annotated subset into 1168 training images and 1000 testing images. We also translate the Cityscapes [44] data set into a thermal data set and use it to learn parameters of our network for effective training. Note that the original data set contains 5000 images in total, including 2975 training images, 500 validation images, and 1525 test images. We take all translated thermal images from this data set as a part of our training set. The details of our SODA against other popular data sets are shown in Table I.

V. Performance Evaluation

Our network is trained on the platform of NVIDIA GeForce GTX TITAN XP GPU with 12-GB memory, and the implementation is based on the public platform PyTorch [55].

A. Parameter Setting

In the generation of thermal images, we train the pix2pixHD [12] from scratch, randomly initializing the weights from a Gaussian distribution with the mean of 0 and the standard deviation of 0.02. The epochs, batch size, and learning rate are set to 20, 1, and 0.0002, respectively.

For the network branch of semantic segmentation, we use the "poly" learning rate policy [11] where current learning rate equals to the base one multiplying $(1-(\text{iter}/(\text{max_iter})))^{\text{power}}$. We set the base learning rate to 0.01 and the power to 0.9. In addition, we set the iteration number, momentum, and weight decay to 100 epochs, 0.9, and 0.0001, respectively. The batch size is set to 14, and the size of input images is scaled to 480×480. To avoid overfitting in training, we adopt several techniques of data augmentation, including mirroring, resizing between 0.5 and 2, and cropping. For the branch of the HED, we first initialize its parameters using the pretrained VGG-16-Net model on the ImageNet data set and then fine-tuned it using the BSD500 data set [37], where all hyperparameters are the same with the setting in [36].

B. Evaluation Metric

We use two standard evaluation metrics, including the pixAcc and the mean Intersection of Union (mIoU). Note that we use the VOC-like evaluation server that calculates mIoU considering the background as one of the categories.

C. Overall Performance

We first compare the proposed approach with the state-of-the-art methods on our SODA data set.

We report the results with ten state-of-the-art methods, including U-Net [15], DFANet [20], FCN-16s [13], FCN-32s [13], DT-EdgeNet [22], ERFNet [52], GCN [53], DUC [17], Deeplabv2 [11], DeepLabv3 [1], and DFN [19] on SODA in Table II. Herein, we utilize the ResNet50 network as the backbone and apply GFT layers in conv2_x. For DFANet and GCN, we reimplemented their methods and fine-tuned their models with the same setting as ours. For other methods, we directly use their released codes.

One can observe that our method achieves 63.64% in mean IoU, better than DFN [53] with 1.21% gain and DeepLabv3 [1] with 2.44% gain, which are the second and third best methods. This superior performance achieved by our method demonstrates the effectiveness of the proposed EC-CNN. We further report per-class IoU on SODA to verify the class-based

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TABLE II
PERFORMANCE COMPARISON IN TERMS OF PER-CLASS IOU WITH 11 METHODS ON THE SODA TEST

	person	building	tree	road	pole	grass	door	table	chair	car	bicycle	lamp	monitor	trafficCone	trash can	animal	fence	sky	river	sidewalk	mIoU
U-Net[15]	57.52	45.17	64.92	69.61	20.69	16.88	4.83	0	19.79	65.19	35.99	0	27.21	31.93	2.97	11.67	12.09	82.05	25.02	10.41	30.86
DFANet[20]	47.88	59.65	62.22	70.59	0.61	32.6	35.94	0.2	27.2	48.73	27.55	17.81	44.4	4.42	25.48	30.93	43.06	81.19	62.27	33.85	38.18
FCN-32s[13]	62.61	60.98	66.26	70.8	17.25	28.92	4.74	0.62	34.24	70.15	43.16	0.27	41.4	39.53	42.74	55.06	45.65	66.09	43.26	42.42	41.91
FCN-16s[13]	63.55	60.72	64.65	72.2	19.84	36.91	8.1	2.36	35.22	72	43.6	25.6	43.47	46	48.22	54.45	54.39	59.51	45.67	40.68	44.7
DT-EdgeNet [22]	42.28	64.25	64.07	70.76	12.76	39.85	28	21.1	37.89	61.88	38.26	37.36	45.66	19.23	51.11	40.54	48.37	72.01	63.59	43.95	45.6
ERFNet[52]	69.34	70.14	70.24	76.12	29.59	43.26	15.66	0	33.68	75.04	48.69	24.51	54.22	38.65	43.47	48.78	47.8	84.93	63.3	42.55	49.11
GCN[53]	71.81	71.03	70.26	76.11	25.53	38.88	36.07	21.27	46.93	79.27	51.41	45.51	62.86	54.97	58.05	59.53	58.34	80.59	72.56	45	56.15
DUC[17]	75.2	73.89	73.3	79.56	40.03	49.49	37.95	17.44	46.3	80.24	54.06	63.45	60.31	59.42	59.68	61.01	59.83	79.63	69.75	50.19	59.2
DeepLabv2[11]	72.97	75.52	72.36	79.72	37.71	50.48	39.74	22.84	43.96	77.05	51.65	60.48	64.38	59.01	62.63	66.87	65.68	85.12	81.1	51.3	60.24
DeepLabv3[1]	71.04	75.14	72.93	79.24	36.42	50.1	40.78	17.38	48.61	57.94	81.95	57.73	65.71	59.65	69.47	69.24	62.58	86.1	79.82	48.98	61.2
DFN[19]	76.92	75.57	74.53	82.2	34.38	51.31	40.52	9.37	45.24	83.98	59.73	73.48	62.42	62.94	65.56	74.37	59.01	88.52	82.76	53.19	62.43
EC-CNN	75.81	76.19	75.27	81.57	41.4	52.37	45.9	20.48	50.56	84.14	57.48	79.2	57.6	61.54	70.57	69.72	64.12	86.79	80.89	52.79	63.64

TABLE III
RESULTS OF THE VARIANTS OF OUR METHOD GFT (CONV2_X+50)

Method	Pixel Acc(%)	Mean IoU(%)
Baseline(50)	79.9	62.3
FiLM(conv2_x+50)	80.1	62.7
SFT(conv2_x+50)	81.4	63
GFT(conv2_x+50)	82.1	63.6

TABLE IV
RESULTS ON DIFFERENT NUMBER OF GFT BLOCKS. THE SHADOW LAYERS HAVE BETTER RESULTS

model	conv2_x	conv3_x	conv4_x	Pixel Acc(%)	Mean IoU(%)
Baseline:DeepLabv3					
50				79.9	62.3
50	✓			82.1	63.6
50		✓		81.6	62.1
50			✓	78.7	51.2
50	✓	✓		83.5	60.1
50		✓	✓	77.9	48.0
50	✓		✓	79.3	50.6
50	✓	✓	✓	78.2	49.7
101				81.6	63.0
101	✓			83.8	63.7
101		✓		84.1	63.9
101			✓	77.1	47.8
101	✓	✓		83.5	61.0
101		✓	✓	70.9	38.1
101	✓		✓	78.8	53.0
101	✓	✓	✓	71.7	40.7

performance of EC-CNN, as presented in Table II, and we achieve the best performance on a majority of classes.

D. Internal Analysis

1) Ablation Study: We first investigate the effectiveness of the proposed GFT layers. To this end, we design three variants of our EC-CNN: 1) Baseline that removes all GFT layers in the EC-CNN; 2) FiLM that replaces GFT layers with FiLM layers [28]; and 3) SFT that removes the gate operation in the GFT layers. The results are shown in Table III. Comparing with baseline, FiLM obtains an improvement of 0.2% (from 79.9% $\rightarrow$ 80.1%) in Pixel Acc and 0.4% (from 62.3% $\rightarrow$ 62.7%) in mIoU. It suggests that the edge information is helpful in improving segmentation performance. However, it ignores the spatial information of conditions. The SFT generates a pair of modulation parameters to apply affine transformation spatially on feature maps of the network. Comparing with FiLM, SFT obtains an improvement of 0.7% (from 80.1% $\rightarrow$ 81.4%) in Pixel Acc and 0.6% (from 62.3% $\rightarrow$ 63%) in mIoU. In addition, we introduce the gate operation in SFT, i.e., GFT, to avoid noisy information of edge maps and obtain an improvement of 0.7% (from 81.4% $\rightarrow$ 82.1%) in Pixel Acc and 0.6% (from 63% $\rightarrow$ 63.6%) in mIoU over SFT.

Then, we also test the performance of GFT insertion in different convolutional blocks, and the results on both ResNet50 and ResNet101 backbones are reported in Table IV. We can observe that the GFT layers are inserted in shadower blocks, and the better results are achieved. For example, the results of conv2_x+50 are the best. The performances of conv2_x+50 and conv3_x+50 are better than conv2_x+50 and conv4_x, and both of them are better than conv3_x+50 and conv4_x+50. Similar observations are obtained on the ResNet101 backbone. It can be explained by the fact that the edge prior information is low-level features, and incorporating them into shadow blocks would lead to a better embedding in thermal image semantic segmentation.

2) Impact of Different Edge Detection Methods: The thermal images are usually with low resolution and low contrast. The edge detection methods could focus on accurate extraction of edge information, which provides very helpful cues to guide thermal image semantic segmentation. Therefore, we integrate the edge detection module to improve performance.

We test the other two state-of-the-art networks of edge detection, including RCF [56] and BDCN [57], in our framework and present the results in Table V. The results show that the better the edge generates, the better the segmentation results achieved.

Besides, we employ the linear transformation-based contrast enhancement followed by edge detection to improve the performance and find that the results change a little, as shown in Table V. It shows that the contrast enhancement operation does not bring the performance gains in our framework.

3) Quality of Thermal Image Generation: To evaluate the visual quality of thermal data in a synthetic subset, we evaluate the visual satisfaction of the translated thermal images. Some examples are shown in Fig. 8, where the results of pix2pix [35] and the corresponding ground truths are presented. We can observe that the pix2pixHD technique [12] adopted in our

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Fig. 8. Results of two image translation methods: pix2pix and pix2pixHD. The first row is the input images, the second row is the results generated by pix2pix, the third row is the results generated by pix2pixHD, and the finally row is the ground truth.

TABLE V
RESULTS OF OUR EC-CNN USING DIFFERENT EDGE DETECTORS

Edge detection Network	Pixel Acc(%)	Mean IoU(%)
HED	82.1	63.6
RCF	82	64
BDCN	82.2	64.5
HED (with contrast enhancement)	82	63.8
RCF (with contrast enhancement)	82.1	64
BDCN (with contrast enhancement)	82.2	64

work can generate better thermal images from RGB ones. It validates the feasibility of the integration of the generated thermal images in our training set.

4) Importance of Synthetic Thermal Data Set: To further justify the effectiveness of our synthetic thermal data, we design different data set configurations in the training stage. Note that we train EC-CNN using the real subset after pretraining in the following configurations: 1) random initialization, that randomly initializes network parameters in EC-CNN; 2) TIR Cityscapes model that only uses the synthetic Cityscapes thermal subset to pretrain EC-CNN; TIR NYUv2 model that only uses the synthetic NYUv2 thermal subset to pretrain EC-CNN; and 4) ImageNet + Gray Cityscapes model that first employs the ImageNet data set to pretrain EC-CNN and then use the grayscale Cityscpaes subset to pretrain. 5) ImageNet model that only uses the ImageNet data set to pretrain EC-CNN; 6) ImageNet + TIR Cityscapes thermal model that first employs the ImageNet data set to pretrain EC-CNN and then uses the synthetic Cityscapes thermal subset to pretrain; 7) ImageNet + TIR NYUv2 thermal model that first employs the ImageNet data set to pretrain EC-CNN and then use the synthetic NYUv2 thermal subset to pretrain; and 8) ImageNet + TIR NYUv2 + TIR Cityscapes model that first employs the ImageNet data set to pretrain EC-CNN and then uses both synthetic NYUv2 thermal subset and synthetic Cityscpaes thermal subset to pretrain. The evaluation results are reported in the Table VI.

The use of the converted data set to train our network significantly improves the performance compared with the random initialization of network parameters. Especially, the mIoU of EC-CNN rises from 25.1% to 50.7% when using the TIR Cityscapes model and from 25.1% to 44.4% when using the TIR NYUv2 model. However, it is not enough to train the EC-CNN well, and thus, we introduce the ImageNet data set to provide a good pertained model. ImageNet model attains 62.1% in mIoU better than the TIR Cityscapes model and the TIR NYUv2 model.

Although the real subset and synthetic subset are in different distributions, they have some common classes, such as person, car, and building. The synthetic data set could enhance the diversity of these classes in the real one and, thus, can improve the accuracy of segmentation. The TIR Cityscapes + ImageNet model attains 63.6% in mIoU that is 1.5% better

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TABLE VI
COMPARISON OF DIFFERENT INITIALIZED MODELS ON EC-CNN

Method	building	tree	table	chair	mIoU(%)
random initialization	30.45	25.29	0	4.65	25.1
TIR Cityscapes model	70.61	71.12	0	11.53	50.7
TIR NYUv2 model	67.21	68.51	0	29.15	44.4
ImageNet+Gray Cityscapes model	75.04	74.45	11.48	40.56	61.5
ImageNet model	75.13	72.92	17.38	48.61	62.1
ImageNet + TIR Cityscapes model	77.57	75.52	17.68	46.61	63.6
ImageNet + TIR NYUv2 model	75.96	73.44	25.23	51.75	63.5
ImageNet + TIR NYUv2 + TIR Cityscapes model	76.19	75.27	20.46	50.56	63.8

than ImageNet model 62.1% in mIoU. The proposed training scheme improves the performance well.

Considering the performance of the added indoor data set, we utilize the TIR NYUv2. The TIR NYUv2 + ImageNet model achieves 63.5% in mIoU that is similar to the TIR Cityscapes + ImageNet model attains 63.6% in mIoU. From the results, we can also observe that the use of the added indoor data set can promote the performance, which is comparable to the use of the outdoor data set Cityscapes in terms of mIoU. However, the mIoU values of per class are different. In particular, for NYUv2, the mIoU values of indoor classes are higher, but the mIoU values of outdoor classes are lower compared with the use of Cityscapes.

We explore the impact of a different order of subsets in progressive learning. There are two reasons why we use the synthetic subset first and then the real one to train our network. First, the synthetic and real subsets are in different distributions, and the real one is more similar to the test set. Second, these two subsets have different classes, and the output has the same number of classes as the real subset. We conduct an extra experiment to validate our design as follows. We change the order of the synthetic and real subsets in progressive learning, i.e., using the real subset first and then the synthetic one to train our network. The IoU results in the classes of person, building, road, pole, car, bicycle, fence, sky, and sidewalk are 46.70, 32.30, 37.82, 28.91, 52.99, 23.96, 24.94, 30.91, and 4.01, respectively, where only the common classes are showed as the output classes are same with the synthetic subset. The results show that the performance of our design achieves clearly better results.

We use both synthetic thermal data set NYUv2 and Cityscapes to train EC-CNN, and the mIoU increases a little, as shown in Table VI. We also add more examples to help understand the importance of the requirement of the new data set, as shown in Fig. 9. We can observe that if only using ImageNet data set to train our model, a number of regions cannot be segmented well since the contents and properties of thermal images are significantly different from natural RGB images. Through using extra data sets, our network could learn good temperature distributions of different classes, and the results are, thus, improved greatly, such as tiny cars, road, and sidewalk in Fig. 9.

The thermal images can be viewed as one kind of gray images, but their contents are totally different. To clarify their differences, we conduct an extra experiment that only uses

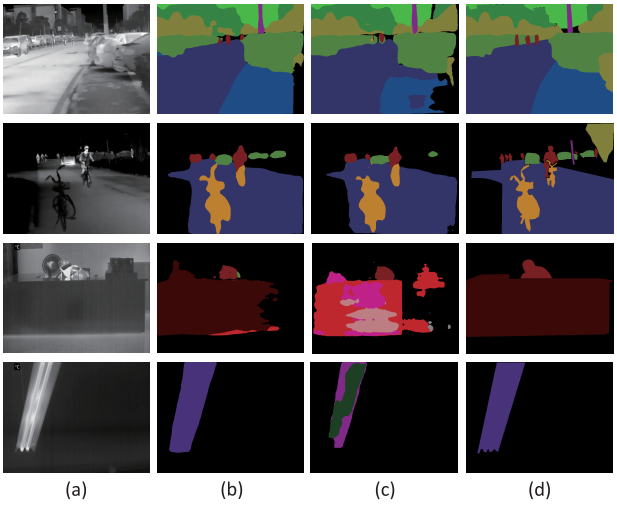


Fig. 9. Effectiveness of additional data set offering in training stage. (a) Input thermal images. (b) Results of both synthetic thermal Cityscapes and synthetic thermal NYUv2 used after ImageNet pretraining. (c) Results of ImageNet pretraining only. (d) Ground truths.

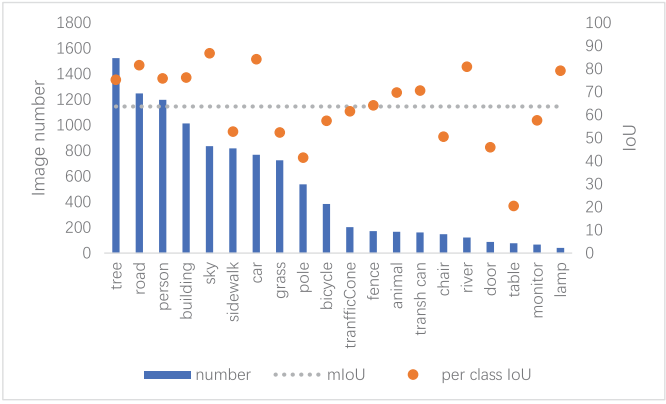


Fig. 10. Sample distribution on all semantic labels on the SODA data set and their corresponding performance in terms of mIoU.

gray images of Cityscapes to train our model and find that the performance degrades, which attains 61.5% in mIoU, as shown in Table VI. While we use the synthetic thermal Cityscapes data set to train our model, the results are improved.

5) Effect of Imbalance Proportion of SODA Data Set: We also list the proportion of per class in the SODA data set and report the performance of EC-CNN in Fig. 10. Most of the classes with a sufficient amount of training data can achieve better mIoU values than others, such as tree, road, person, and building. If the number of training samples is small, but the objects are usually with high contrast with background or other objects, these classes can also achieve high mIoU values, such as river and lamp. If the number of training samples is small and the objects are easily disturbed by a thermal crossover with other subjects, the performance of these classes drastically decreases, such as pole and table.

6) Analysis of Running Speed: In addition, we evaluate the running speed of the proposed framework. As shown in Table VII, the EC-CNN (ResNet50) takes about 0.13–0.17 s per image on average on GeForce GTX TITAN XP GPU,

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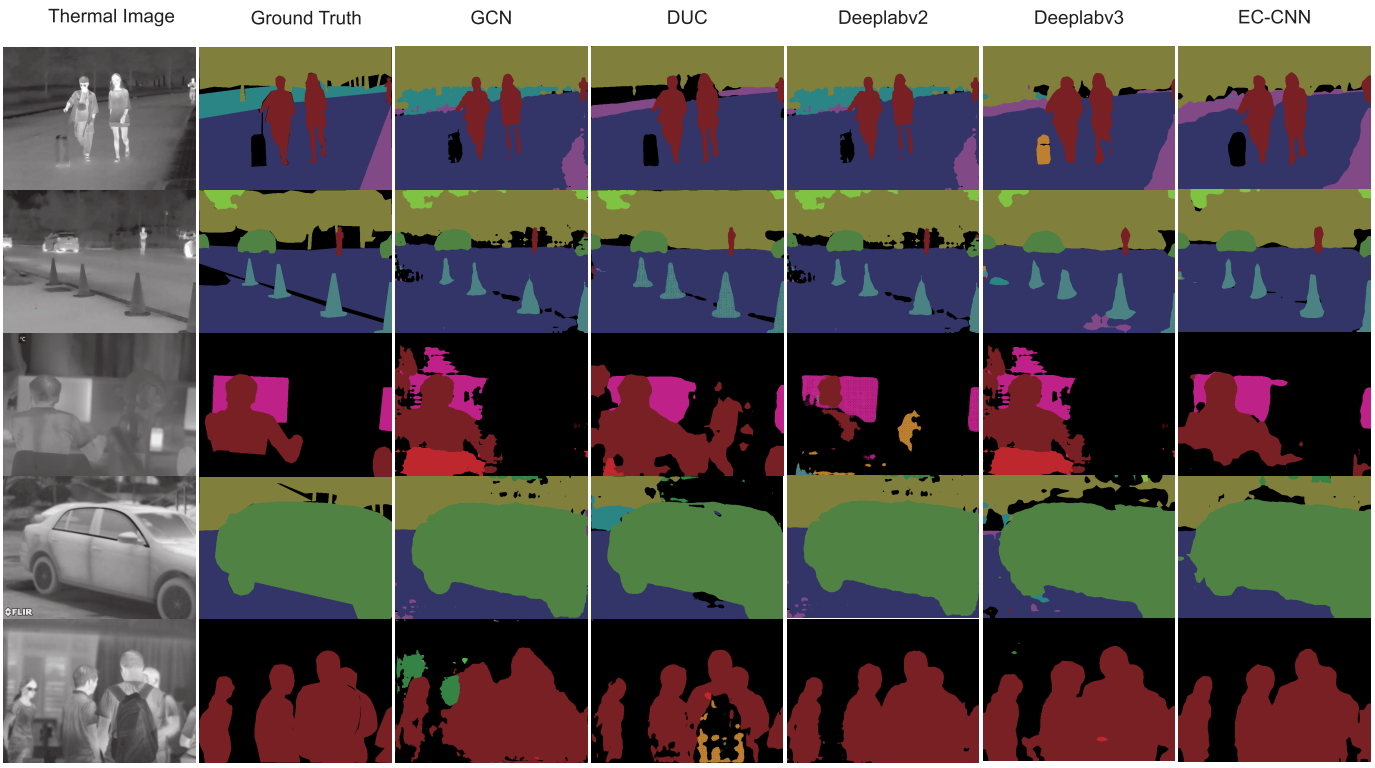


Fig. 11. Examples of thermal semantic segmentation results on SODA.

TABLE VII

SPEED GROWS WITH DEEPER NETWORKS, WHERE THE NUMBERS IN THE BRACKETS REFER TO THE DEPTHS OF RESNET AND "CONV2_X" INDICATES THAT WE INSERT THE EDGE GUIDANCE IN CONV2_X LAYER OF EC-CNN

Method	time
DeepLabv3(50)	0.11
EC-CNN(conv2_x+50)	0.13
EC-CNN(conv3_x+50)	0.14
EC-CNN(conv4_x+50)	0.17
DeepLabv3(101)	0.18
EC-CNN(conv2_x+101)	0.20
EC-CNN(conv3_x+101)	0.21
EC-CNN(conv4_x+101)	0.45

which are near to the speed of the baseline DeepLabv3. Similar results are also observed on the backbone ResNet101. It suggests that our framework can achieve clear improvements in both pixAcc and mIoU over the baseline DeepLabv3 with a modest impact on the running speed.

7) Performance on RGB Image Data Set: The proposed method is also suitable for image semantic segmentation task. This is because the tasks of image semantic segmentation and thermal image semantic segmentation have the same goal that is to classify each pixel. However, the task of thermal image semantic segmentation has some different challenges and properties. For example, the thermal images usually have low resolution and low contrast. We integrate the edge guidance module to address these challenges, and our method is thus more suitable for thermal image semantic segmentation than image semantic segmentation.

TABLE VIII

RESULTS OF OUR EC-CNN WITH OTHER METHODS ON THE PASCAL VOC DATA SET

Method	Pixel Acc(%)	Mean IoU(%)
GCN	92.8	70.1
DUC	93.1	71.9
DeepLabv3	93.5	73.6
DFN	93.6	74.2
EC-CNN(conv2_x+50)	93.8	74.5

To validate our explanations, we test our network on the RGB image segmentation data set, PASCAL-VOC. In a specific, we use 2703 images in PASCAL-VOC 2007 and 2012 data sets for training and 210 for testing. Table VIII shows the performance of our model compared with the following state-of-the-art methods. From the results, we can observe that our EC-CNN achieves the best. For example, we achieve 74.5% in mIoU, which is 0.9% better than DeepLabv3 and 0.3% better than DFN. In other words, the performance gains are not obvious as in thermal image semantic segmentation, as shown in Table III. The results demonstrate our observation that our EC-CNN is more suitable for thermal image semantic segmentation.

8) Qualitative Results: The qualitative comparison is shown in Fig. 11. As can be observed from these visual results, our method outputs more precise predictions than the other four methods despite the existence of low resolution and contrast, ambiguous object boundaries caused by thermal crossover, and imaging noised introduced by thermal sensors. Taking the third and fifth row for example, our approaches can also successfully handle the ambiguous object boundaries caused

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by thermal crossovers, such as the right arm and TV monitor. These regions with similar textures can be recognized and separated by the guidance from the introduced edge information. In general, through effectively exploiting edge priors, our approach produces more reasonable results of thermal image semantic segmentation in spite of the presence of confusing labels.

VI. Conclusion

In this work, we presented a new data set SODA for a new vision task of thermal image semantic segmentation. SODA contains 2168 images that are richly annotated with 20 semantic labels. Besides, we also provide Cityscapse-based synthetic thermal images to increase the scale of SODA. With the SODA data set, we performed extensive experimental analysis to identify successes and limitations of some leading approaches to the task of thermal image semantic segmentation and proposed a more suitable method for this task. In a particular, we design a novel EC-CNN architecture to explicitly incorporate edge priors. The experimental results demonstrate the effectiveness of the proposed EC-CNN.

From the evaluation results, utilizing edge prior information in the task of thermal image semantic segmentation has many advantages, especially when semantic regions are low resolution and contrast, boundary-ambiguous caused by thermal crossover, and imaging noises introduced by thermal sensors.

In addition, based on evaluation results and observations, we highlight some components that are essential for thermal image semantic segmentation. The first one is the data set. At present, there is no large-scale annotated data set for thermal image semantic segmentation. Although our SODA provides a first data set for this new task, the well-annotated data are not enough, especially for the development of deep learning techniques in the research direction. Therefore, a large-scale well-annotated thermal image data set is essential for boosting the development of thermal image semantic segmentation. Meanwhile, we should also study transfer learning techniques to transform massive knowledge from RGB data into the thermal domain, which alleviates the dependence on large-scale data.

The second one is the method. As discussed earlier, thermal information has special properties comparing with visible spectrum data (i.e., RGB data), such as insensitivity to lighting conditions and thermal crossover. How to design a suitable and powerful network for thermal data is extremely important. We provide one solution by incorporating edge priors in existing RGB semantic segmentation network, and other priors can also be investigated, such as semantic relations and context knowledge. In addition, we also use the advantages of RGB data to mitigate the disadvantages of thermal data, such as thermal crossover using multimodal learning techniques. On one side, the thermal sensors are more effective in capturing objects than visible spectrum cameras under poor lighting conditions and bad weather. On the other side, visible spectrum cameras are more discernible in separating two different subjects under thermal crossover. Therefore, RGB and thermal information complement each other and contribute to semantic segmentation in different aspects together.

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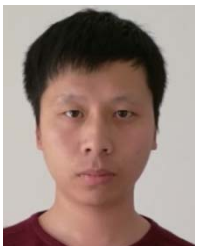
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